Available online at www.sciencedirect.com
SCIENCE DIRECT®
Image and Vision Computing 24 (2006) 926–934
www.elsevier.com/locate/imavis

Image encryption using chaotic logistic map

N.K. Pareek [a,b], Vinod Patidar [a], K.K. Sud [a,b,*]

[a] Department of Physics, College of Science Campus, M.L.S. University, Udaipur 313002, Rajasthan, India
[b] University Computer Centre, Vigyan Bhawan, New Campus, M.L.S. University, Udaipur 313002, Rajasthan, India

Received 10 August 2004; received in revised form 11 August 2005; accepted 6 February 2006

Abstract

In recent years, the chaos based cryptographic algorithms have suggested some new and efficient ways to develop secure image encryption techniques. In this communication, we propose a new approach for image encryption based on chaotic logistic maps in order to meet the requirements of the secure image transfer. In the proposed image encryption scheme, an external secret key of 80-bit and two chaotic logistic maps are employed. The initial conditions for the both logistic maps are derived using the external secret key by providing different weightage to all its bits. Further, in the proposed encryption process, eight different types of operations are used to encrypt the pixels of an image and which one of them will be used for a particular pixel is decided by the outcome of the logistic map. To make the cipher more robust against any attack, the secret key is modified after encrypting each block of sixteen pixels of the image. The results of several experimental, statistical analysis and key sensitivity tests show that the proposed image encryption scheme provides an efficient and secure way for real-time image encryption and transmission.
© 2006 Elsevier B.V. All rights reserved.

Keywords: Image cipher; Chaotic cryptography; Logistic map

1. Introduction

In recent years, owing to frequent flow of digital images across the world over the transmission media, it has become essential to secure them from leakages. Many applications like military image databases, confidential video conferencing, medical imaging system, cable TV, online personal photograph album, etc. require reliable, fast and robust security system to store and transmit digital images. The requirements to fulfill the security needs of digital images have led to the development of good encryption techniques. During the last decade, numerous encryption algorithms [1–13] have been proposed in the literature based on different principles. Among them, chaos based encryption techniques are considered good for practical use as these techniques provide a good combination of speed, high security, complexity, reasonable computational overheads and computational power etc. The digital images have certain characteristics such as: redundancy of data, strong correlation among adjacent pixels, being less sensitive as compared to the text data i.e. a tiny change in the attribute of any pixel of the image does not drastically degrade the quality of the image and bulk capacity of data etc. Consequently, the traditional ciphers like IDEA, AES, DES, RSA etc. are not suitable for real time image encryption as these ciphers require a large computational time and high computing power. For real time image encryption only those ciphers are preferable which take lesser amount of time and at the same time without compromising security. An encryption scheme which runs very slowly, even may have higher degree of security features would be of little practical use for real time processes.

A number of chaos based image encryption scheme have been developed in recent years which we discuss in brief in this paragraph. In 1992, Bourbakis and Alexopoulos [1] have proposed an image encryption scheme which utilizes the SCAN language to encrypt and compress an image simultaneously. Fridrich [4] demonstrated the construction of a symmetric block encryption technique based on two-dimensional standard baker map. There are three basic steps in the method of Fridrich [4]: (a) choose a chaotic map and generalize it by introducing some parameter, (b) discretize the chaotic map to a finite square lattice of points that represent pixels, (c) extend the discretized map to three-dimensions and further compose it with a simple diffusion mechanism. Further, Scharinger [5] designed a chaotic Kolmogorov-flow-based image encryption technique, in which whole image is taken as

* Corresponding author. Address: Department of Physics, College of Science Campus, M.L.S. University, Udaipur 313002, Rajasthan, India. Tel.: +91 294 2418123; fax: +91 294 2413150.
E-mail address: kksud@yahoo.com (K.K. Sud).

0262-8856/$ - see front matter © 2006 Elsevier B.V. All rights reserved.
doi:10.1016/j.imavis.2006.02.021

a single block and which is permuted through a key-controlled chaotic system. In addition, a shift register pseudo random generator is also adopted to introduce the confusion in the data. Yen and Guo [6] proposed an encryption method called BRIE based on chaotic logistic map. The basic principle of BRIE is bit recirculation of pixels, which is controlled by a chaotic pseudo random binary sequence. The secret key of BRIE consists of two integers and an initial condition of the logistic map. Further, Yen and Guo [9] also proposed an encryption method called CKBA (Chaotic Key Based Algorithm) in which a binary sequence as a key is generated using a chaotic system. The image pixels are rearranged according to the generated binary sequence and then XORed and XNORed with the selected key. Later in 2002, Li and Zheng [11] pointed out some defects in the encryption schemes presented in the references [6,9] and also discussed some possible improvements on them. Recently, Li et al. [12] have proposed a video encryption technique based on multiple digital chaotic systems which is known as CVES (Chaotic Video Encryption Scheme). In this scheme, 2^n chaotic maps are used to generate pseudo random signals to mask the video and to perform pseudo random permutation of the masked video. Very recently, Chen et al. [13] have proposed a symmetric image encryption in which a two-dimensional chaotic map is generalized to three-dimension for designing a real time secure image encryption scheme. This approach employs the three-dimensional cat map to shuffle the positions of the image pixels and uses another chaotic map to confuse the relationship between the encrypted and its original image.

The characteristics of the chaotic maps have attracted the attention of cryptographers to develop new encryption algorithms. As these chaotic maps have many fundamental properties such as ergodicity, mixing property and sensitivity to initial condition/system parameter and which can be considered analogous to some cryptographic properties of ideal ciphers such as confusion, diffusion, balance and avalanche property etc. In this communication, a new image encryption scheme is proposed based on chaotic logistic maps in order to meet the requirements of the secure image transfer. Further in the proposed image encryption scheme an external secret key (as used by Chen et al. [13] for image encryption and by Pareek et al. [14,15] for text ciphers) of 80-bit and two chaotic logistic maps are employed. The initial conditions for the both logistic maps are derived using the external secret key by providing different weightage to its bits (see Eq. (6) for details). In the algorithm, the first logistic map is used to generate numbers ranging from 1 to 24 (numbers may be repeated). The initial condition of the second logistic map is modified from the numbers, generated by the first logistic map. By modifying the initial condition of the second logistic map in this way, its dynamics gets further randomized. In the proposed encryption process, eight different types of operations are used to encrypt the pixels of an image and which operation will be used for a particular pixel is decided by the outcome of the second logistic map. Thus, the second chaotic map further increases the confusion in the relationship between the encrypted and its original image. To make the cipher more robust against any attack, after each encryption of a block of sixteen pixels, the secret key is modified. In Section 2, we discuss the step by step procedure of image encryption and in Section 3, the security analysis of the proposed image encryption scheme such as statistical analysis, key and plaintext sensitivity analysis, key space analysis etc. to prove its security against the most common attacks. Finally, in Section 4, we conclude the paper.

2. The proposed image encryption procedure

In this section, we discuss the step by step procedure of the proposed image encryption as well as decryption process using two chaotic logistic maps.

1 The proposed image encryption process utilizes an external secret key of 80-bit long. Further, the secret key is divided into blocks of 8-bit each, referred as session keys.

$$K = k_1 k_2 \ldots k_{20} \text{ (in hexadecimal)}, \tag{1}$$

here, k_i's are the alphanumeric characters (0–9 and A–F) and each group of two alphanumeric characters represents a session key. Alternatively, the secret key can be represented in ASCII mode as

$$K = K_1 K_2 \ldots K_{10} \text{ (in ASCII)}, \tag{2}$$

here, each K_i represents one 8-bit block of the secret key i.e. session key.

2 In the proposed algorithm, two chaotic logistic maps are employed to achieve the goal of image encryption which are as follow:

$$X_{n+1} = 3.9999 X_n (1 - X_n), \tag{3}$$

$$Y_{n+1} = 3.9999 Y_n (1 - Y_n). \tag{4}$$

Throughout the algorithm, we keep the value of the system parameter of the both logistic maps to be constant (i.e. 3.9999) which corresponds to a highly chaotic case while the initial conditions (X_0 and Y_0) for these maps are calculated using some mathematical manipulations on session keys.

3 To calculate the initial condition X_0 for the first logistic map, choose three blocks of session keys i.e. $K_4 K_5 K_6$ and convert them into a binary string as:

$$B_1 = K_{41} K_{42} \ldots K_{48} K_{51} K_{52} \ldots K_{58} K_{61} K_{62} \ldots K_{68}, \tag{5}$$

here, K_{ij}'s are the binary digits (0 or 1) of the ith block of the session key. Next, we compute a real number X_{01} using the above binary representation as:

$$\begin{aligned} X_{01} = (&K_{41} \times 2^0 + K_{42} \times 2^1 + \cdots + K_{48} \times 2^7 + K_{51} \\ &\times 2^8 + K_{52} \times 2^9 + \cdots + K_{58} \times 2^{15} + K_{61} \times 2^{16} \\ &+ K_{62} \times 2^{17} + \cdots + K_{68} \times 2^{23})/2^{24}. \end{aligned} \tag{6}$$

Further, we compute another real number X_{02} as follows

$$X_{02} = \sum_{i=13}^{18} (k_i)_{10}/96, \tag{7}$$

here ki's are parts of secret key in hexadecimal mode as explained in Eq. (1). Now we compute the initial condition X_0 for the first logistic map using X_{01} and X_{02} as:

$$X_0 = (X_{01} + X_{02})\text{mod } 1. \tag{8}$$

4 We generate a sequence of 24 real numbers $f_1, f_2, \ldots, f_{24}$ by iterating the first logistic map using the initial condition obtained in step 3. Keeping in mind that we have considered only those values, which fall in the interval [0.1,0.9], the other values are discarded from the sequence. The real number sequence is converted into an integer sequence using the following formula

$$P_k = \text{int}(23 \times (f_k - 0.1)/0.8) + 1, \tag{9}$$

where $k = 1,2,\ldots,24$.

5 To calculate the initial condition Y_0 for the second logistic map, we choose three blocks of session keys i.e. $K_1K_2K_3$, and convert them into a binary string as:

$$B_2 = K_{11}K_{12}\ldots K_{18}K_{21}K_{22}\ldots K_{28}K_{31}K_{32}\ldots K_{38}, \tag{10}$$

here K_{ij}'s are the binary digits (0 or 1) of the ith block of the session key. We then compute a real number Y_{01} using the above binary representation as:

$$Y_{01} = (B_2)_{10}/2^{24}. \tag{11}$$

Further, we compute another real number Y_{02} as follows

$$Y_{02} = \left(\sum_{k=1}^{24} \underline{B_2[P_k]} \times 2^{k-1}\right)/2^{24}, \tag{12}$$

here, $\underline{B_2[P_k]}$ denotes the value of P_k^{th} bit in the binary string B_2 i.e. it is either 0 or 1. Now we compute the initial condition Y_0 for the second logistic map using Y_{01} and Y_{02} as:

$$Y_0 = (Y_{01} + Y_{02})\text{mod } 1. \tag{13}$$

6 Read the three consecutive bytes from the image file. These three bytes represent the value of the red, green and blue (RGB) color respectively and together form a single pixel of the image.

7 We divide the range [0.1,0.9] into 24 non-overlapping intervals and arrange them into eight different groups. Then we assign different type of operation corresponding to each of these groups. In Table 1, we have given the interval ranges for the groups and their corresponding operations. Further, we iterate once the second logistic map using the initial condition Y_0 obtained in Step 5. The outcome of the second logistic map decides which operation to be performed for encryption/decryption of red, green and blue (RGB) color bytes.

This step is repeated $(K_{10})_{10}$ (i.e. the decimal equivalent of 10th session key) times. Finally, the encrypted bytes of red, green and blue colors are written in a file i.e. encryption of a pixel is achieved.

We repeat steps 6 and 7 for next 15 pixels of the image file.

8 After encryption of a block of 16 pixels of image file, we modify the session keys K_1 to K_9 as follows:

$$(K_i)_{10} = ((K_i)_{10} + (K_{10})_{10})\text{mod } 256, \quad (1 \le i \le 9). \tag{14}$$

Table 1
Different groups of non-overlapping intervals of Y values and corresponding operations for image encryption and decryption

Group no.	Intervals of Y values	Operations for encryption/decryption
1	0.10–0.13, 0.34–0.37, 0.58–0.62	NOT operation, i.e. invert the bits of all three RGB bytes
2	0.13–0.16, 0.37–0.40, 0.62–0.66	$R \oplus K_4$, $G \oplus K_5$ and $B \oplus K_6$
3	0.16–0.19, 0.40–0.43, 0.66–0.70	Encryption $((R)_{10}+(K_4)_{10}+(K_5)_{10})$ mod 256, $((G)_{10}+(K_5)_{10}+(K_6)_{10})$ mod 256, $((B)_{10}+(K_6)_{10}+(K_4)_{10})$ mod 256 Decryption $((R)_{10}+256-(K_4)_{10}-(K_5)_{10})$, $((G)_{10}+256-(K_5)_{10}-(K_6)_{10})$, $((B)_{10}+256-(K_6)_{10}-(K_4)_{10})$
4	0.19–0.22, 0.43–0.46, 0.70–0.74	Encryption $NOT(R \oplus K_4)$, $NOT(G \oplus K_5)$, $NOT(B \oplus K_6)$ Decryption $(NOT(R)) \oplus K_4$, $(NOT(G)) \oplus K_5$, $(NOT(B)) \oplus K_6$
5	0.22–0.25, 0.46–0.49, 0.74–0.78	Similar to Group 2 except that K_7, K_8 and K_9 are used in lieu of K_4, K_5 and K_6, respectively.
6	0.25–0.28, 0.49–0.52, 0.78–0.82	Similar to Group 3 except that K_7, K_8 and K_9 are used in lieu of K_4, K_5 and K_6, respectively.
7	0.28–0.31, 0.52–0.55, 0.82–0.86	Similar to Group 4 except that K_7, K_8 and K_9 are used in lieu of K_4, K_5 and K_6, respectively.
8	0.31–0.34, 0.55–0.58, 0.86–0.90	No operations are made on R,G and B bytes

Here, K_i's are the session keys, R, G and B, respectively, are the red, green and blue levels of image pixels.

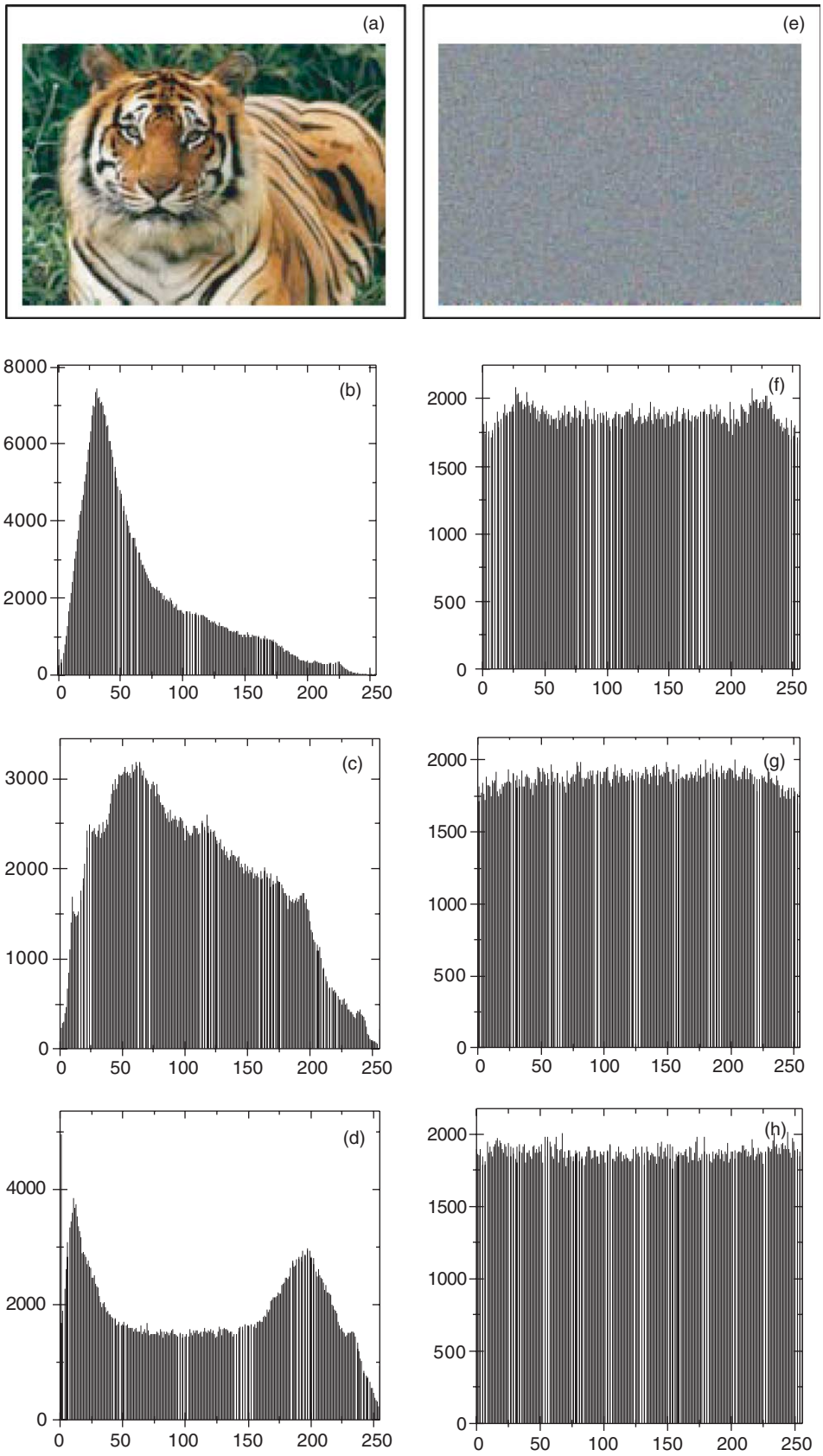

Fig. 1. Histogram analysis: Frame (a) shows a plain image. Frames (b), (c) and (d) respectively, show the histograms of red, green and blue channels of the plain image shown in Frame (a). Frame (e) shows the encrypted image of the plain image shown in Frame (a) using the secret key '*ABCDEF0123456789FF05*' (in hexadecimal). Frames (f), (g) and (h) respectively, show the histograms of red, green and blue channels of the encrypted image shown in Frame (e).

After modifying the secret key in the above manner and by taking the last value of X from the step 4 as a initial condition for the first logistic map (in other words $X_0 = f_{24}$), we again generate a sequence of 24 real numbers as explained in the step 4 and then repeat steps 5–7 until the entire image file is exhausted.

The process of decryption is completely similar to the encryption process described above; only the difference would be in the step 7 i.e. in the case of encryption process, we iterate the second logistic map $(K_{10})_{10}$ times and do the respective operations for encryption (which depend on the outcome of the logistic map) after each iteration of the map. However, in the case of decryption process, we first iterate the second logistic map $(K_{10})_{10}$ times and do the respective operations for decryption (which again depends on the outcome of the logistic map) in the reverse order.

3. Security analysis

A good encryption procedure should be robust against all kinds of cryptanalytic, statistical and brute-force attacks. In this section, we discuss the security analysis of the proposed image encryption scheme such as statistical analysis, sensitivity analysis with respect to the key and plaintext, key space analysis etc. to prove that the proposed cryptosystem is secure against the most common attacks.

3.1. Statistical analysis

It is well known that many ciphers have been successfully analyzed with the help of statistical analysis and several statistical attacks have been devised on them. Therefore, an ideal cipher should be robust against any statistical attack. To prove the robustness of the proposed image encryption procedure, we have performed statistical analysis by calculating the histograms, the correlations of two adjacent pixels in the encrypted images and the correlation coefficient for several images and its corresponding encrypted images of an image database.

(a) *Histogram analysis*: An image-histogram illustrates how pixels in an image are distributed by graphing the number of pixels at each color intensity level. We have calculated and analyzed the histograms of the several encrypted as well as its original colored images that have widely different content. One example of such histogram analysis is shown in Fig. 1. Particularly in Frame (a), we have shown the original image and in Frames (b), (c) and (d) respectively, the histograms of red, blue and green channels of the original image (Frame (a)). In Frame (e), we have shown the encrypted image of the original image (Frame (a)) using the secret key '*ABCDEF0123456789FF05*' (in hexadecimal) and in Frames (f), (g) and (h) respectively, the histograms of red, blue and green channels of the encrypted image (Frame (e)). It is clear from Fig. 1 that the histograms of the encrypted image are fairly uniform and significantly different from the respective

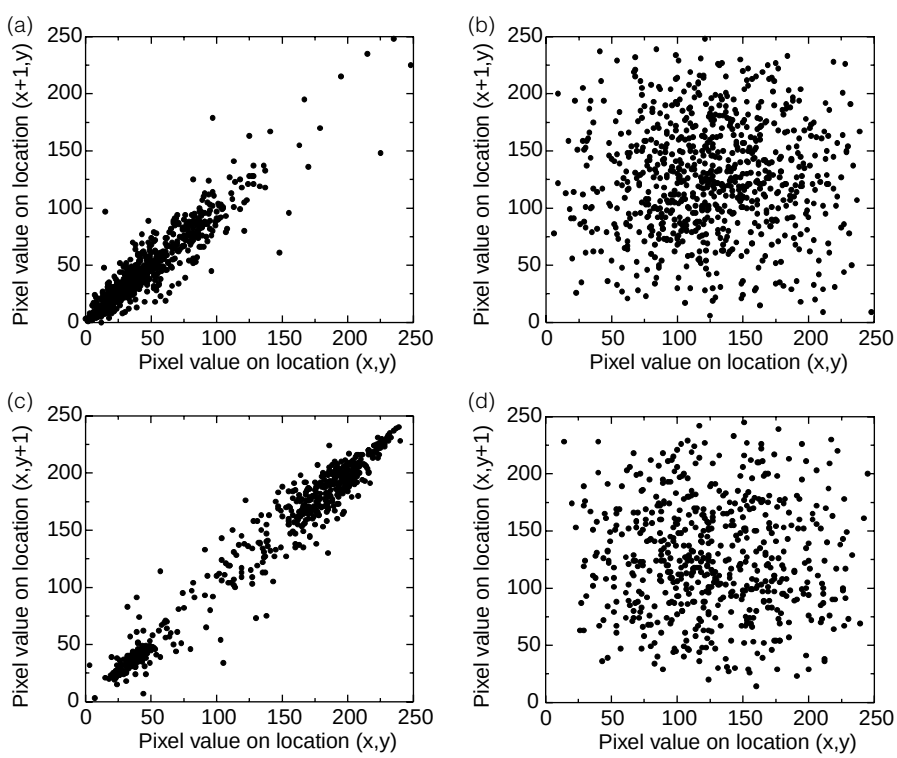

Fig. 2. Correlation of two adjacent pixels: Frames (a) and (b) respectively, show the distribution of two horizontally adjacent pixels in the plain and encrypted images shown in Fig. 1a and 1e. Frames (c) and (d) respectively, show the distribution of two vertically adjacent pixels in the plain and encrypted images shown in Fig. 1a and 1e.

histograms of the original image and hence does not provide any clue to employ any statistical attack on the proposed image encryption procedure.

(b) *Correlation coefficient analysis*: In addition to the histogram analysis, we have also analyzed the correlation between two vertically as well as horizontally adjacent pixels in the several images and their encrypted images. In Fig. 2, we have shown the distribution of two adjacent pixels in the original and encrypted images shown in Fig. 1a and 1e. Particularly, in Frames (a) and (b), we have depicted the distributions of two horizontally adjacent pixels in the original and encrypted images respectively. Similarly, in Frames (c) and (d) respectively, the distributions of two vertically adjacent pixels in the original and encrypted images have been depicted. Moreover, we have also calculated the correlation between two vertically as well as horizontally adjacent pixels in the original encrypted images. For this calculation, we have used the following formula

$$C_r = \frac{N\sum_{j=1}^{N}(x_j \times y_j) - \sum_{j=1}^{N} x_j \times \sum_{j=1}^{N} y_j}{\sqrt{\left(N\sum_{j=1}^{N} x_j^2 - \left(\sum_{j=1}^{N} x_j\right)^2\right) \times \left(N\sum_{j=1}^{N} y_j^2 - \left(\sum_{j=1}^{N} y_j\right)^2\right)}}, \tag{15}$$

where x and y are the value of two adjacent pixels in the image and N is total number of pixels selected from the image for the calculation. In the Table 2, we have given the correlation coefficients for the original and encrypted images shown in Fig. 1a and 1e respectively. It is clear from the Fig. 2 and Table 2 that there is negligible correlation between the two adjacent pixels in the encrypted image. However, the two adjacent pixels in the original image are highly correlated.

We have also done extensive study of the correlation between image and its corresponding encrypted image by using the proposed encryption algorithm. We have used the USC-SIPI image database which is a collection of digitized images available and maintained by the University of Southern California primarily to support research in image processing, image analysis, and machine vision. The database is divided into four different categories based on the basic character of the pictures. Currently, four volumes available at USC-SIPI site are—textures, aerials, miscellaneous and sequences. We have chosen miscellaneous volume to measure the correlation coefficient of USC-SIPI image database (freely available at http://sipi.usc.edu/database/). The miscellaneous volume consists of 44 images out of which 16 are colored and 28 monochrome. The secret key '*A98FDAB5632C10FFE805*' has been used for encryption process. Results are shown in the Table 3. The average correlation coefficient is very small which implies that no correlation exist between original and its corresponding cipher images.

Table 2
Correlation coefficients for the two adjacent pixels in the original and encrypted images shown in Fig. 1

	Original image (Fig. 1a)	Encrypted image (Fig. 1e)
Horizontal	0.8710	0.0041
Vertical	0.4668	−0.0337

3.2. Sensitivity analysis

An ideal image encryption procedure should be sensitive with respect to the secret key i.e. the change of a single bit in

Table 3
Correlation coefficient between the image and corresponding cipher image for a number of images of the USC-SIPI image database. The encryption has been done using the secret key '*A98FDAB5632C10FFE805*'.

File name	File description	Size	Type	Correlation coefficient
4.1.01	Girl	256×256	Color	−0.003961
4.1.02	Couple	256×256	Color	−0.008119
4.1.03	Girl	256×256	Color	−0.005314
4.1.04	Girl	256×256	Color	−0.011863
4.1.05	House	256×256	Color	−0.009243
4.1.06	Tree	256×256	Color	−0.003994
4.1.07	Jelly beans	256×256	Color	0.001905
4.1.08	Jelly beans	256×256	Color	0.002922
4.2.01	Splash	512×512	Color	−0.007686
4.2.02	Girl (Tiffany)	512×512	Color	−0.008184
4.2.03	Baboon	512×512	Color	−0.007076
4.2.04	Girl (Lenna)	512×512	Color	0.002325
4.2.05	Airplane (F-16)	512×512	Color	−0.002833
4.2.06	Sailboat on lake	512×512	Color	−0.008145
4.2.07	Peppers	512×512	Color	0.000815
House	House	512×512	Color	−0.010095
5.1.09	Moon surface	256×256	Gray	0.000779
5.1.10	Aerial	256×256	Gray	−0.007672
5.1.11	Airplane	256×256	Gray	−0.004110
5.1.12	Clock	256×256	Gray	−0.011780
5.1.13	Resolution chart	256×256	Gray	−0.017896
5.1.14	Chemical plant	256×256	Gray	−0.008989
5.2.08	Couple	512×512	Gray	−0.006210
5.2.09	Aerial	512×512	Gray	−0.006024
5.2.10	Stream and bridge	512×512	Gray	−0.001512
7.1.01	Truck	512×512	Gray	−0.000297
7.1.02	Airplane	512×512	Gray	−0.001944
7.1.03	Tank	512×512	Gray	−0.006857
7.1.04	Car and APCs	512×512	Gray	−0.006561
7.1.05	Truck and APCs	512×512	Gray	−0.011244
7.1.06	Truck and APCs	512×512	Gray	−0.002139
7.1.07	Tank	512×512	Gray	−0.006582
7.1.08	APC	512×512	Gray	0.001338
7.1.09	Tank	512×512	Gray	−0.002992
7.1.10	Car and APCs	512×512	Gray	0.006279
boat.512	Fishing Boat	512×512	Gray	−0.010011
elaine.512	Girl (Elaine)	512×512	Gray	−0.006344
gray21.512	21 level step wedge	512×512	Gray	−0.004317
numbers.512	256 level test pattern	512×512	Gray	−0.009823
ruler.512	Pixel ruler	512×512	Gray	−0.007839
5.3.01	Man	1024×1024	Gray	−0.012766
5.3.02	Airport	1024×1024	Gray	−0.011103
7.2.01	Airplane	1024×1024	Gray	0.003126
testpat.1k	General test pattern	1024×1024	Gray	−0.000720

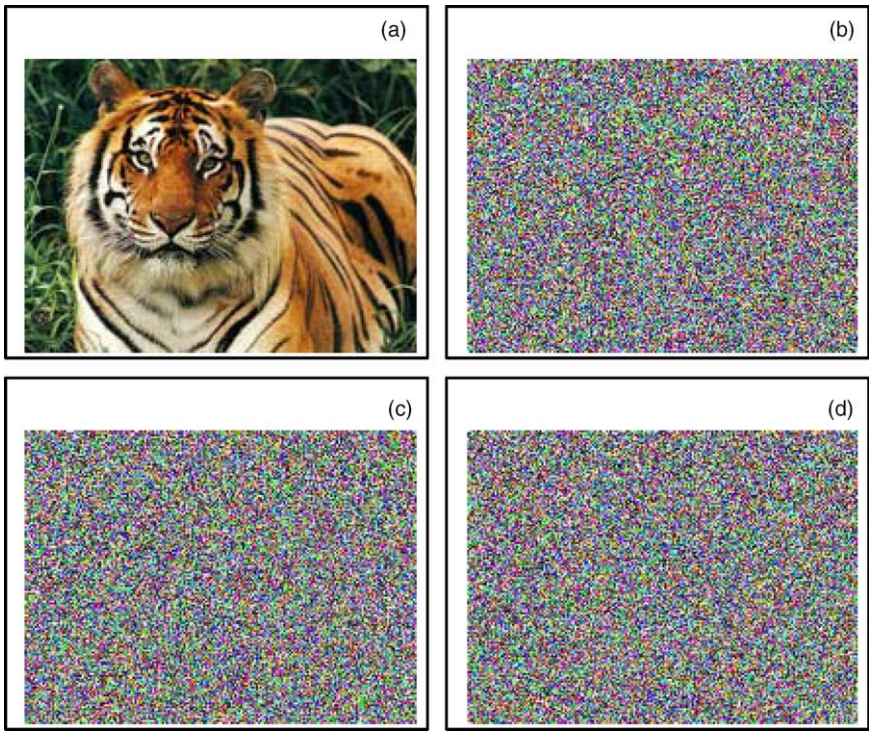

Fig. 3. Key sensitivity test I: Frame (a) shows a plain image, Frames (b), (c) and (d) respectively, show the encrypted images of the plain image shown in Frame (a) using the secret keys '*A148CB3FD50766E47405*', '*B148CB3FD50766E47405*' and '*A148CB3FD50766E47406*'(all in hexadecimal).

the secret key should produce a completely different encrypted image. For testing the key sensitivity of the proposed image encryption procedure, we have performed the following steps:

(a) An original image (Fig. 3a) is encrypted by using the secret key '*A148CB3FD50766E47405*' (in hexadecimal) and the resultant image is referred as encrypted image A (Fig. 3b).
(b) The same original image is encrypted by making the slight modification in the secret key i.e. '*B148CB3FD50766E47405*' (the most significant bit is changed in the secret key) and the resultant image is referred as encrypted image B (Fig. 3c).
(c) Again, the same original image is encrypted by making the slight modification in the secret key i.e. '*A148CB3FD50766E47406*' (the least significant bit is changed in the secret key) and the resultant image is referred as encrypted image C (Fig. 3d).
(d) Finally, the three encrypted images A, B and C are compared.

In Fig. 3, we have shown the original image as well as the three encrypted images produced in the aforesaid steps. It is not easy to compare the encrypted images by simply observing these images. So for comparison, we have calculated the correlation between the corresponding pixels of the three encrypted images. For this calculation, we have used the same formula as given in Eq. (15) except that in this case x and y are the values of corresponding pixels in the two encrypted images to be compared. In Table 4, we have given the results of the correlation coefficients between the corresponding pixels of the three encrypted images A, B and C. It is clear from the table that no correlation exists among three encrypted images even though these have been produced by using slightly different secret keys.

Moreover, in Fig. 4, we have shown the results of some attempts to decrypt an encrypted image with slightly different secret keys than the one used for the encryption of the original image. Particularly, in Frames (a) and (b) respectively, the original image and the encrypted image produced using the secret key 'A039FD52FC87CD1E4406' (in hexadecimal) are shown whereas in Frames (c) and (d) respectively, the images after the decryption of the encrypted image (shown in Frame (b)) with the secret keys 'A139FD52FC87CD1E4406' (in hexadecimal) and 'A039FD52FD87CD1E4406' (in hexadecimal). It is clear that the decryption with a slightly different key

Table 4
Correlation coefficients between the corresponding pixels of the three different encrypted images obtained by using slightly different secret key of an image shown in Fig. 3

Image 1	Image 2	Correlation coefficient
Encrypted image A (Fig. 3b)	Encrypted image B (Fig. 3c)	0.00393
Encrypted image B (Fig. 3c)	Encrypted image C (Fig. 3d)	−0.00627
Encrypted image C (Fig. 3d)	Encrypted image A (Fig. 3b)	0.00289

Fig. 4. Key sensitivity test II: Frames (a) and (b) respectively, show a plain image and its encrypted image using the secret key '*A039FD52FC87CD1E4406*' (in hexadecimal). Frames (c) and (d) respectively, show the images after the decryption of the encrypted image shown in Frame (b) using the secret keys '*A139FD52FC87CD1E4406*' (in hexadecimal) and '*A039FD52FD87CD1E4406*' (in hexadecimal).

fails completely and hence the proposed image encryption procedure is highly key sensitive.

We have also measured the number of pixels change rate (NPCR) to see the influence of changing a single pixel in the original image on the encrypted image by the proposed algorithm. The NPCR measure the percentage of different pixel numbers between the two images. We take two encrypted images, C_1 and C_2, whose corresponding original images have only one-pixel difference. We define a two-dimensional array D, having the same size as the image C_1/C_2. The $D(i,j)$ is determined from $C_1(i,j)$ and $C_2(i,j)$. If $C_1(i,j)=C_2(i,j)$ then $D(i,j)=1$ otherwise $D(i,j)=0$. The NPCR is defined by the following equation [13].

$$\mathrm{NPCR} = \frac{\sum_{i,j} D(i,j)}{wh} \times 100\%, \tag{16}$$

where w and h are the width and height of encrypted image. We obtained NPCR for a large number of images by using our encryption scheme and found it to be over 99% showing thereby that the encryption scheme is very sensitive with respect to small changes in the plaintext.

3.3. Key space analysis

For a secure image cipher, the key space should be large enough to make the brute force attack infeasible. The proposed image cipher has $2^{80}(\approx 1.20893\times 10^{24})$ different combinations of the secret key. An image cipher with such a long key space is sufficient for reliable practical use. However, one can have longer key for image encryption/-decryption in the proposed image cipher and it can be easily incorporated in the algorithm by making slight modification. For this purpose, we have to reduce interval size shorter than the one shown in the Table 1. A longer key would require more computational time for encryption/decryption which may not be preferable for real time transmission.

In the proposed image cipher, two chaotic maps are employed for encryption/decryption which are sensitive on the initial condition. The initial conditions for these two logistic maps are calculated from the secret key with different formulae. Due to limitation of numerical precision of a computer, 24-bit are used to assemble a floating-point number, i.e. initial condition. Moreover, the secret key is modified, after encryption/decryption of every block of 16 pixels, from the part of the session key, i.e. K_{10} and further, the initial

Table 5
Average ciphering speed of a few different sized colored images

Image size (in pixels)	Bits/pixels	Average encryption/ decryption time (s)
256×256	24	0.33–0.39
512×512	24	0.38–0.40
1024×1024	24	6.26–6.32
2048×2048	24	25.15–25.32

conditions for these two chaotic logistic maps are recalculated from the modified secret key.

3.4. Time analysis

Apart from the security consideration, running speed of the algorithm is also an important aspect for a good encryption algorithm. We have measured the encryption/decryption rate of several colored images of different sized by using the proposed image encryption scheme. The time analysis has been done on Pentium-4 with 256 MB RAM computer. The average encryption/decryption time taken by the algorithm for different sized images are shown in the Table 5.

4. Conclusions

In this communication, a new way of image encryption scheme have been proposed which utilizes two chaotic logistic maps and an external key of 80-bit. The initial conditions for both the logistic maps are derived using the external secret key by providing weightage to its bits corresponding to their position in the key. In the proposed encryption process, eight different types of operations are used to encrypt the pixels of an image and which operation will be used for a particular pixel is decided by the outcome of the logistic map. To make the cipher more robust against any attack, the secret key is modified after encrypting a block of sixteen pixels of the image. We have carried out statistical analysis, key sensitivity analysis and key space analysis to demonstrate the security of the new image encryption procedure. Finally, we conclude with the remark that the proposed method is expected to be useful for real time image encryption and transmission applications.

Acknowledgements

The support from the special assistance programmed of the University Grants Commission (UGC), New Delhi to the Department of Physics, M.L.S. University is gratefully acknowledged. One of us (VP) also acknowledges to the University Grants Commission (UGC), New Delhi for providing senior research fellowship.

References

[1] N. Bourbakis, C. Alexopoulos, Picture data encryption using SCAN pattern, Pattern Recogn. 25 (1992) 567–581.
[2] Refregier, B Javidi, Optical image encryption based on input plane and fourier plane random encoding, Opt. Lett. 20 (1995) 767–769.
[3] H.K.L. Chang, J.L. Liu, A linear quad tree compression scheme for image encryption, Signal Process. 10 (4) (1997) 279–290.
[4] Fridrich Jiri, Symmetric ciphers based on two dimensional chaotic maps, Int. J. Bifurcat Chaos 8 (6) (1998) 1259–1284.
[5] J. Scharinger, Fast encryption of image data using chaotic Kolmogrov flow, J. Electronic Eng 7 (2) (1998) 318–325.
[6] J.C. Yen, J.I. Guo, A new image encryption algorithm and its VLSI architecture, in: Proceedings of the IEEE workshop signal processing systems, 1999, pp. 430–437
[7] J.C. Yen, J.I. Guo, An efficient hierarchical chaotic image encryption algorithm and its VLSI realization, IEE Proc. Vis. Image Process. 147 (2000) 167–175.
[8] H. Cheng, X.B. Li, Partial encryption of compressed image and videos, IEEE Trans. Signal Process. 48 (8) (2000) 2439–2451.
[9] J.C. Yen, J.I. Guo, A new chaotic key based design for image encryption and decryption, Proceedings of the IEEE International Symposium Circuits and Systems, vol. 4, 2000, pp. 49–52
[10] C.C. Chang, M.S. Hwang, T.S. Chen, A new encryption algorithm for image cryptosystems, J. Syst. Software 58 (2001) 83–91.
[11] S. Li, X. Zheng, Cryptanalysis of a chaotic image encryption method, in: Proceedings of the IEEE International. symposium on circuits and systems, Scottsdale, AZ, USA, 2002
[12] S. Li, X. Zheng, X. Mou, Y. Cai, Chaotic encryption scheme for real time digital video, Proceedings of the SPIE on electronic imaging, San Jose, CA, USA, 2002
[13] G. Chen, Y. Mao, C.K. Chui, A symmetric image encryption based on 3D chaotic maps, Chaos Solitons Fractals 21 (2004) 749–761.
[14] N.K. Pareek, Vinod Patidar, K.K. Sud, Discrete chaotic cryptography using external key, Phys. Lett. A 309 (2003) 75–82.
[15] N.K. Pareek, Vinod Patidar, K.K. Sud, Cryptography using multiple one-dimensional chaotic maps, Commun. Nonlinear Sci. Numer. Simul. 10 (7) (2005) 715–723.